TECHNICAL APPENDIX

requirements.txt and src/fairness_metrics.py

1. requirements.txt

The following package list is intended for installing the notebook's Python dependencies. Exact package versions are not pinned in the supplied notebook.

```
numpy
pandas
matplotlib
seaborn
scikit-learn
catboost
jupyter
ipykernel
nbformat
tabulate
```

Install with: pip install -r requirements.txt

2. src/fairness_metrics.py

The reusable module below computes per-group selection rates and true-positive rates, followed by max-minus-min gaps. It is a standalone utility file and should be saved as src/fairness_metrics.py.

```
import numpy as np
import pandas as pd


def calculate_fairness_metrics(y_true, y_pred, groups, attribute_name):
    """Calculate selection rate and true positive rate for each group."""
    data = pd.DataFrame({
        "actual": np.asarray(y_true),
        "predicted": np.asarray(y_pred),
        "group": np.asarray(groups)
    })

    results = []
    for group_name, group_data in data.groupby("group", dropna=False):
        n = len(group_data)
        actual_positive = (group_data["actual"] == 1).sum()
        predicted_positive = (group_data["predicted"] == 1).sum()
        selection_rate = predicted_positive / n

        if actual_positive > 0:
            true_positives = (
                (group_data["actual"] == 1) &
                (group_data["predicted"] == 1)
            ).sum()
            tpr = true_positives / actual_positive
        else:
            tpr = np.nan

        results.append({
            "Attribute": attribute_name,
            "Group": group_name,
            "Group Size": n,
            "Actual Positives": actual_positive,
            "Predicted Positives": predicted_positive,
            "Selection Rate": selection_rate,
            "TPR": tpr
        })

    return pd.DataFrame(results)


def calculate_fairness_gaps(fairness_df):
```

```
    """Return max-minus-min gaps for selection rate and TPR."""
    return {
        "Selection Rate Gap": (
            fairness_df["Selection Rate"].max()
            - fairness_df["Selection Rate"].min()
        ),
        "TPR Gap": (
            fairness_df["TPR"].max()
            - fairness_df["TPR"].min()
        )
    }
```

3. Important implementation note

The supplied notebook already defines a fairness-metric function in its own cells. This module is a reusable extraction for a repository structure; it does not automatically refactor the notebook to import the module. To use it from the notebook, add the project root to the Python import path as needed and import the functions from src.fairness_metrics.